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CryptoFace: End-to-End Encrypted Face Recognition

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Abstract

Face recognition is central to many authentication, security, and personalized applications. Yet, it suffers from significant privacy risks, particularly arising from unauthorized access to sensitive biometric data. This paper introduces CryptoFace, the first end-to-end encrypted face recognition system with fully homomorphic encryption (FHE). It enables secure processing of facial data across all stages of a face-recognition process—feature extraction, storage, and matching—without exposing raw images or features. We introduce a mixture of shallow patch convolutional networks to support higher-dimensional tensors via patch-based processing while reducing the multiplicative depth and, thus, inference latency. Parallel FHE evaluation of these networks ensures near-resolution-independent latency. On standard face recognition benchmarks, CryptoFace significantly accelerates inference and increases verification accuracy compared to the state-of-the-art FHE neural networks adapted for face recognition. CryptoFace will facilitate secure face recognition systems requiring robust and provable security. The code is available at https://github.com/human-analysis/CryptoFace.

1. Introduction

Face Recognition (FR) [13, 26, 44] has become integral to identity management in many practical applications, from unlocking personal devices to facilitating law enforcement and accessing financial services. These systems process sensitive biometric data that, if compromised, can lead to privacy invasions, identity theft, and unauthorized surveillance. Unlike passwords, biometric data is immutable—once compromised, it cannot be changed, which elevates the need for robust security mechanisms to protect it. Such protections are also mandated by legal regulations on the acquisition, storage, and usage of biometric data [34], *e.g.* the European Union's General Data Protection Regulation (GDPR) [43]. FR systems in the wild consist of three entities: a probe face image, a feature extractor (*i.e.* a FR neural network), and a reference database of face features.

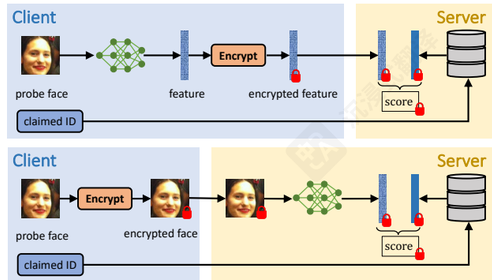

Figure 1. Secure FR systems. (1) Top: existing secure FR system which encrypts only the features, rather than raw face images, limits practical utility and security; (2) bottom: proposed end-to-end encrypted face recognition system which ensures stronger protection of the user's face image and feature while safeguarding the server's reference database.

The feature extractor processes a face image to generate its corresponding compact feature representation.

Secure FR Systems. FR systems meet increasing security challenges. An adversarial client could attempt to infer biometric data from the server's reference database or leverage the feature extractor to generate adversarial probes to deceive the system. Similarly, an adversarial server might exploit client-provided data to extract biometric information or infer sensitive attributes such as gender, age, or race. Existing secure FR systems apply homomorphic encryption (HE) in a client-server two-party scenario [3, 15] to ensure security of *face features*. HE allows computation over encrypted data and provides provable post-quantum security [7, 9, 16, 17, 32]. As shown in Figure 1 (top), the client's probe feature and the reference database at a server are encrypted. The client performs feature extraction locally and cannot delegate this task to the server. Verification is performed by the server directly within the encrypted domain. Limiting security measures to *feature protection only* reduces the practical utility of such systems. Moreover, such a secure FR system is vulnerable to a template recovery attack [2] where an adversarial client could attempt to infer features in the server's reference database. Unlike HE,

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CryptoFace：端到端加密人脸识别

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摘要

人脸识别是许多身份验证、安全和个性化应用的核心。然而，它面临着严重的隐私风险，特别是来自未经授权访问敏感生物特征数据的威胁。本文介绍了*CryptoFace*，这是第一个采用全同态加密（*FHE*）技术的端到端加密人脸识别系统。它能够在人脸识别过程的各个阶段——特征提取、存储和匹配——安全地处理面部数据，而不会暴露原始图像或特征。我们引入了一种混合浅层卷积网络的组合，通过基于块的处理来支持高维张量，同时减少乘法深度，从而降低推理延迟。这些网络的并行*FHE*评估确保了接近分辨率无关的延迟。在标准人脸识别基准测试中，*CryptoFace*与为人脸识别适配的最先进的*FHE*神经网络相比，显著加速了推理并提高了验证精度。*CryptoFace*将促进需要强大且可证明安全性的安全人脸识别系统。代码可在*https://github.com/human-analysis/CryptoFace*获取。

1. 简介

人脸识别（FR） [13, 26, 44] 已成为许多实际应用中身份管理的核心，从解锁个人设备到协助执法和访问金融服务。这些系统处理敏感的生物特征数据，如果遭到泄露，可能导致隐私侵犯、身份盗窃和未经授权的监控。与密码不同，生物特征数据是不可更改的——一旦泄露，就无法更改，这提高了保护它的强大安全机制的需求。此类保护措施也受到关于生物特征数据获取、存储和使用的法律规定的约束 [34]，例如欧盟的通用数据保护条例（GDPR） [43]。现实世界中的人脸识别系统由三个实体组成：一个探测人脸图像、一个特征提取器（即人脸识别神经网络）以及一个包含人脸特征参考数据库。

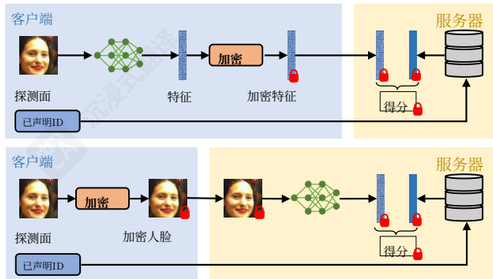

图1. 安全人脸识别系统。 (1) 顶部：现有的安全人脸识别系统仅加密特征，而非原始人脸图像，限制了其实际效用和安全性； (2) 底部：提出的端到端加密人脸识别系统，确保用户人脸图像和特征得到更强保护，同时保护服务器参考数据库。

特征提取器处理人脸图像以生成其对应的紧凑特征表示。

保障 FR 系统。FR 系统面临日益严峻的安全挑战。一个对抗性客户端可能会尝试从服务器的参考数据库中推断生物特征数据，或利用特征提取器生成对抗性探针来欺骗系统。类似地，一个对抗性服务器可能会利用客户端提供的数据来提取生物特征信息，或推断性别、年龄或种族等敏感属性。现有的安全 FR 系统在客户端-服务器两方场景中应用同态加密 (HE) [3, 15] 来确保人脸特征的安全性。HE 允许在加密数据上进行计算，并提供可证明的后量子安全性 [7, 9, 16, 17, 32]。如图 1（上）所示，客户端的探针特征和服务器处的参考数据库均被加密。客户端在本地执行特征提取，无法将此任务委托给服务器。验证由服务器在加密域内直接执行。仅将安全措施限于特征保护会降低此类系统的实际效用。此外，这种安全 FR 系统容易受到模板恢复攻击 [2]，其中对抗性客户端可能会尝试推断服务器参考数据库中的特征。与 HE 不同的是，

which provides robust and provable security guarantees, other lines of research in privacy-preserving face recognition focus on different security or privacy issues, such as adversarial attacks [12], viewing attacks [35], and information leakage [23].

CryptoFace. To address security vulnerabilities in existing secure FR systems, we introduce CryptoFace, the *first end-to-end encrypted* FR system using fully homomorphic encryption (FHE). All operations–including feature extraction, feature matching, and score comparison–are performed entirely within the encrypted domain without decryption at any point. As shown in Figure 1 (bottom), the client encrypts the probe face image and sends it to the server. The server uses a neural network (NN) to extract an encrypted feature. This encrypted feature is matched against the encrypted reference feature stored in the server's database, producing an encrypted similarity score. The score is compared against a threshold, and the encrypted match/no-match result is returned to the client, who alone can decrypt it. By securing the user's sensitive biometric data throughout the verification process and protecting the server's reference database, CryptoFace offers robust end-to-end security for FR systems.

Realizing this goal presents three major technical challenges: (1) Homomorphic evaluation of state-of-the-art (SoTA) convolutional neural networks (CNNs) is computationally demanding due to the high multiplicative depth of CNNs [1, 7, 9, 28]. (2) Although several approaches [1, 18, 28] have demonstrated homomorphic evaluation of CNNs on low-resolution images, they cannot directly process higher-resolution face images. (3) FR requires the cosine similarity measure [13], which cannot be directly computed on FHE. Existing secure FR [3, 15] circumvented the evaluation of the cosine function under FHE by normalizing the features before encryption.

CryptoFaceNet is designed to address these technical challenges. A face image is divided into a grid of non-overlapping patches [14], and each is processed independently by a shallow patch CNN (PCNN). The mixture of PCNNs is jointly trained to learn the inter-patch relationships. Due to the lower resolution of individual patch, we reduce the multiplicative depth required for each PCNN. We also optimize convolutional blocks to further minimize the multiplicative depth and adopt other recent advancements for efficient FHE convolution [28] and low-degree polynomial [39] activation functions. Under FHE, the mixture of PCNNs is evaluated in parallel and features are additively aggregated, significantly accelerating the feature extraction process. CryptoFaceNet scales effectively to high-resolution face images and maintains near-resolution-independent latency due to parallelism. Additionally, we design a distribution-aware low-degree polynomial approximation of the cosine similarity function to efficiently compute the similarity score under FHE.

We evaluate CryptoFace on standard FR benchmarks, comparing its performance to SoTA FHE CNNs [1, 28] adapted for FR. Our results show that CryptoFace not only improves the one-to-one verification accuracy by up to $+8.8\%$ but also speeds up the encrypted FR by $7\times$. CryptoFace supports arbitrary-resolution images, maintaining near-constant latency across different resolutions. We summarize the **contributions** of this paper below:

1. **End-to-End Encrypted Face Recognition:** CryptoFace is the first secure FR system to perform feature extraction, feature matching, and score thresholding entirely within the encrypted domain, eliminating the need for decryption at any stage. CryptoFace enhances security and expands the applicability of secure FR.
2. **Efficient and Scalable Architecture:** CryptoFaceNet is a novel FHE-compatible architecture that reduces computational overhead by minimizing multiplicative depth and is scalable to high-resolution face images.
3. **Feasibility and Efficacy Demonstration:** We present the first practical implementation of end-to-end encrypted face recognition, demonstrating its feasibility on standard FR benchmarks under FHE.

2. Background and Related Work

2.1. Homomorphic Encryption (HE)

Homomorphic encryption (HE) is a class of encryption schemes that are considered quantum-secure and enable computations on encrypted data without requiring decryption. HE schemes are based on the Learning with Errors (LWE) problem [16, 17] or Ring Learning with Errors (RLWE) [32]. Among different HE schemes, the Cheon-Kim-Kim-Song (CKKS) encryption scheme [7–9] is particularly well-suited for encrypted inference in neural networks since it supports fixed-point approximate arithmetic over complex and real numbers.

Encryption and Decryption. A *cleartext* message vector $\mu \in \mathbb{C}^{\frac{N}{2}}$ is first encoded into a *plaintext* message m, which is subsequently encrypted into a *ciphertext* $\boldsymbol{c}$ using a public key $pk = (-\langle a, sk\rangle + e, a)$. Here, $\langle \cdot, \cdot \rangle$ is dot product operator, a is a random ring, e is an encryption noise, and sk is the secret key. The encryption and decryption processes are defined as follows:

$$\begin{aligned} \text{Encrypt}(m, pk) &= (m, 0) + pk = (c_0, c_1) = \boldsymbol{c} \\ \text{Decrypt}(\boldsymbol{c}, sk) &= c_0 + \langle c_1, sk\rangle = m + e \end{aligned} \tag{1}$$

The CKKS scheme uses a residue cyclotomic polynomial ring $\mathcal{R}_{Q_\ell} = \mathbb{Z}_{Q_\ell}[X]/(X^N+1)$ to encode cleartext vectors. The modulus is defined as $Q_\ell = \prod_{i=0}^{\ell} q_\ell, 0 \le \ell \le L$. The polynomial degree N determines the message capacity, allowing $\frac{N}{2}$ complex numbers to be packed into $\frac{N}{2}$ slots.

它提供了强大且可证明的安全保证，其他隐私保护人脸识别研究则关注不同的安全或隐私问题，例如对抗性攻击 [12], 、观察攻击 [35], 和信息泄露 [23]。

CryptoFace。为解决现有安全人脸识别（FR）系统中的安全漏洞，我们引入了CryptoFace，这是首个采用全同态加密（FHE）技术的端到端加密FR系统。所有操作——包括特征提取、特征匹配和分数比较——均在加密域内完成，全程无需解密。如图1（底部）所示，客户端加密待测人脸图像并发送到服务器。服务器使用神经网络（NN）提取加密特征。该加密特征与服务器数据库中存储的加密参考特征进行匹配，生成加密相似度分数。该分数与阈值进行比较，并将加密的匹配/不匹配结果返回给客户端，只有客户端才能解密。通过在整个验证过程中保护用户的敏感生物特征数据，并保护服务器的参考数据库，CryptoFace为FR系统提供了强大的端到端安全性。

实现这一目标面临三大主要技术挑战：(1) 由于卷积神经网络（CNN）具有高乘法深度，对最先进（SoTA）CNN进行同态评估计算量巨大 [1, 7, 9, 28]。(2) 尽管已有几种方法 [1, 18,28] 在低分辨率图像上实现了CNN的同态评估，但它们无法直接处理高分辨率人脸图像。(3) FR需要余弦相似度度量 [13], ，而该度量在同态加密（FHE）上无法直接计算。现有的安全FR [3, 15] 通过加密前对特征进行归一化来绕过在同态加密下计算余弦函数的问题。

CryptoFaceNet 旨在解决这些技术挑战。人脸图像被划分为一个由不重叠的块组成的网格 [14], ，每个块由浅层块 CNN（PCNN）独立处理。PCNN 的混合体被联合训练以学习块间关系。由于单个块的分辨率较低，我们减少了每个 PCNN 所需的乘法深度。我们还优化了卷积块以进一步最小化乘法深度，并采用其他最近的进展来实现高效的 FHE 卷积 [28] 和低度多项式 [39] 激活函数。在 FHE 下，PCNN 的混合体并行评估，特征被加性聚合，显著加速了特征提取过程。CryptoFaceNet 可有效扩展到高分辨率人脸图像，并由于并行性而保持近分辨率无关的延迟。此外，我们设计了一种分布感知的低度多项式余弦相似度函数近似方法，以高效地计算 FHE 下的相似度分数。

计算 FHE 下的相似度分数。

我们在标准的FR基准测试上评估了CryptoFace，将其性能与为FR适配的SoTA FHE CNNs [1, 28]进行比较。我们的结果表明，CryptoFace不仅将一对一验证准确率提高了高达 $+8.8\%$，还加快了加密FR的速度 $7\times$。CryptoFace支持任意分辨率的图像，在不同分辨率下保持近乎恒定的延迟。我们总结本文的贡献如下：

1. 端到端加密人脸识别：CryptoFace是首个在加密域内完成特征提取、特征匹配和分数阈值化的安全FR系统，无需在任何阶段进行解密。CryptoFace增强了安全性，并扩展了安全FR的适用范围。

2. 高效且可扩展的架构：CryptoFaceNet是一种新型FHE兼容架构，通过最小化乘法深度来降低计算开销，并可以扩展到高分辨率人脸图像。

3. 可行性与有效性验证：我们首次展示了端到端加密人脸识别的实用实现，在FHE下于标准FR基准测试中验证了其可行性。

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2. 背景与相关工作

2.1. 同态加密 (HE)

同态加密 (HE) 是一类被认为具有量子安全性的加密方案，能够在不要求解密的情况下对加密数据进行计算。HE 方案基于学习错误 (LWE) 问题 [16, 17]或环学习错误 (RLWE) [32]。在多种 HE 方案中，Cheon-Kim-Kim-Song (CKKS) 加密方案 [7–9] 特别适用于神经网络中的加密推理，因为它支持复数和实数的定点近似运算。

加密与解密。明文消息向量 $\mu \in \mathbb{C}^{\frac{N}{2}}$ 首先被编码为明文消息 $\overline{m}$，随后使用公钥 $pk = (-\langle a, sk\rangle + e, a)$ 将其加密为密文 $\boldsymbol{c}$。这里，$\langle \cdot, \cdot \rangle$ 是点积运算符，a 是一个随机环，e 是加密噪声，sk 是密钥。加密和解密过程定义如下：

$$\begin{aligned} \text{Encrypt}(m, pk) &= (m, 0) + pk = (c_0, c_1) = \boldsymbol{c} \\ \text{Decrypt}(\boldsymbol{c}, sk) &= c_0 + \langle c_1, sk\rangle = m + e \end{aligned} \tag{1}$$

CKKS方案使用余数循环多项式环 $\mathcal{R}_{Q_\ell} = \mathbb{Z}_{Q_\ell}[X]/(X^N+1)$ 对明文向量进行编码。模数定义为 $Q_\ell = \prod_{i=0}^{\ell} q_\ell \ 0 \le \ell \le L$，。多项式度数 N 决定了消息容量，允许 $\frac{N}{2}$ 复数被打包进 $\frac{N}{2}$ 个槽位。

Supported Operations. The CKKS scheme supports two homomorphic operations—addition and multiplication—and one automorphic operation, rotation. These operations are defined as follows [7, 28]:

$$\begin{aligned}
&\underbrace{[m_1]\oplus[m_2]}_{\text{ciphertext-ciphertext}} \approx \underbrace{[m_1]\oplus m_2}_{\text{ciphertext-plaintext}} \approx [\underbrace{m_1+m_2}_{\text{element-wise add}}] \\
&\underbrace{[m_1]\otimes[m_2]}_{\text{ciphertext-ciphertext}} \approx \underbrace{[m_1]\otimes m_2}_{\text{ciphertext-plaintext}} \approx [\underbrace{m_1\times m_2}_{\text{element-wise mul}}] \\
&\mathrm{Rot}([m],r)=[\mathrm{Rot}(m,r)]
\end{aligned} \tag{2}$$

Here, $[\cdot]$ represents an encrypted message or vector and $\mathrm{Rot}(\cdot)$ denotes a left cyclical rotation of the vector by r positions. Homomorphic addition ($\oplus$) and multiplication ($\otimes$) can be applied between two ciphertexts or between a ciphertext and plaintext, enabling element-wise computations.

Multiplicative Level and Depth. Each ciphertext is associated with a level ℓ, an integer indicating the number of homomorphic multiplications that can be performed before decryption fails. A function with a multiplicative depth k–defined as the number of sequential homomorphic multiplications it involves—consumes k levels. After each multiplication, the polynomial modulus Q_ℓ must be rescaled, transitioning from Q_ℓ to $Q_{\ell-1}$ to maintain the scale [7].

LHE and FHE. CKKS without bootstrapping is a leveled homomorphic encryption (LHE) scheme, allowing a limited number of multiplications determined by the initial level L of a freshly encrypted ciphertext. To evaluate functions with arbitrary depth, fully homomorphic encryption (FHE) incorporates a bootstrapping operation [4, 8, 30] to *refresh* ciphertexts, effectively resetting their level to enable further computation. For deeper neural networks, bootstrapping must be periodically applied to prevent decryption failures. However, this process is computationally expensive, with high latency due to the large number of rotation operations involved. Additionally, bootstrapping has a significant memory footprint because of the large size of the bootstrapping operators. While a freshly encrypted ciphertext starts with L levels, bootstrapping reduces the available levels to $L-K$, as K levels are consumed during the evaluation of polynomials required for bootstrapping.

Computational Complexity. Bootstrapping is slower than rotation or ciphertext-ciphertext multiplication by two orders of magnitude [25]. Ciphertext-ciphertext multiplication is slower than ciphertext-plaintext multiplication or addition by two orders of magnitude [25].

2.2. Homomorphic CNNs (HCNNs)

Homomorphic CNNs (HCNNs) are CNNs that are compatible with the operations that HE supports in Equation 2. We categorize HCNNs into FHENets and LHENets depending on whether bootstrapping is used or not, respectively. LHENets include CryptoNets [18], LoLa [5], Faster CryptoNets [11], while recent FHENets include MPCNN [28] and AutoFHE [1]. FHENets achieve SoTA prediction accuracy on image classification datasets but with much higher latency than LHENets. Existing HCNNs to speed up the inference on encrypted images have focused on two aspects, packing for convolutions and polynomial activation.

Packing for Convolutions refers to efficiently packing three-dimensional tensors to reduce the complexity of HE multiplication and rotation for convolutional layers [24, 28]. MPCNN designs multiplexed convolution by integrating (1) repeated packing and (2) multiplexed packing [28]. (1) A ciphertext with the cyclotomic polynomial degree N can pack $\frac{N}{2}$ numbers. Given a vector $x\in\mathbb{R}^d$ with $d<\frac{N}{2}$, a repeated vector is $x^{(M)}=[x,x,\cdots,x]$ with $M=\lfloor\frac{N}{2d}\rfloor$ copies of x. $x^{(M)}$ is encrypted to fill out all slots. The repeated packing can accelerate convolution and bootstrapping operations. A larger M leads to faster inference. (2) When the convolutional stride exceeds 1, gaps between valid values are introduced, causing some ciphertext slots to remain unused. MPCNN addresses this issue by packing numbers from different channels into alternate slots, effectively filling these gaps and fully utilizing all ciphertext slots, thereby preventing sparsity. Additionally, channels are computed in parallel to speed up convolutional layers.

Polynomial Activation. HCNNs cannot employ ReLU since it is not a homomorphism. So, they adopt polynomial activations (e.g., monomial, Chebyshev or Hermite polynomials) to replace ReLU. Monomial polynomials are widely used since they allow HCNNs to be formulated as traditional polynomial networks. Examples include CryptoNets [18], LoLa [5], and Faster CryptoNets [11]. However, training with monomial polynomials often becomes unstable due to exploding gradients. Minimax approximations using Chebyshev polynomials can achieve high-precision approximations of ReLU functions [27–29], but their high polynomial degree leads to prohibitively large multiplicative depths. AESPA [39] addresses this issue by introducing low-degree Hermite polynomials to reduce the multiplicative depth and proposes basis-wise normalization to stabilize training. Furthermore, search-based AutoFHE [1] explores layer-wise mixed-degree polynomials to further decrease multiplicative depth.

CryptoFaceNet Modules. We build upon the above mentioned prior advances for accelerating HCNN inference, specifically adopting multiplexed convolution [28] and low-degree, basis-wise normalized Hermite activation [39]. Our primary focus, however, lies in addressing other outstanding challenges of end-to-end encrypted FR, including processing high-resolution encrypted images and minimizing the multiplicative depth to reduce computationally expensive bootstrapping operations. We overcome these challenges through CryptoFaceNet, a novel architecture design.

支持的运算。CKKS方案支持两种同态运算——加法和乘法——以及一种自同态运算，旋转。这些运算定义如下 [7, 28]:

$$\begin{aligned}
&\underbrace{[m_1]\oplus[m_2]}_{\text{ciphertext-ciphertext}} \approx \underbrace{[m_1]\oplus m_2}_{\text{ciphertext-plaintext}} \approx [\underbrace{m_1+m_2}_{\text{element-wise add}}] \\
&\underbrace{[m_1]\otimes[m_2]}_{\text{ciphertext-ciphertext}} \approx \underbrace{[m_1]\otimes m_2}_{\text{ciphertext-plaintext}} \approx [\underbrace{m_1\times m_2}_{\text{element-wise mul}}] \\
&\mathrm{Rot}([m],r)=[\mathrm{Rot}(m,r)]
\end{aligned} \tag{2}$$

这里， $[\cdot]$ 表示一个加密消息或向量，Rot(·)表示向量向左循环旋转 r 个位置。同态加法($\oplus$)和乘法($\otimes$)可以应用于两个密文之间，或者密文和明文之间，从而实现逐元素计算。

乘法级别和深度。每个密文都与一个级别 ℓ相关联，这是一个整数，表示在解密失败之前可以执行的同态乘法次数。一个具有乘法深度k（定义为它涉及的连续同态乘法次数）的函数消耗 k 个级别。每次乘法后，多项式模数 Q_ℓ 必须重新缩放，从 Q_ℓ 过渡到 $Q_{\ell-1}$ 以保持规模 [7]。

LHE和FHE。没有重置操作的CKKS是一种分层同态加密（LHE）方案，允许有限数量的乘法，这些乘法次数由新加密密文的初始级别 L决定。为了评估任意深度的函数，全同态加密（FHE）包含一个重置操作 [4, 8, 30]来刷新密文，有效地将它们的级别重置为以进行进一步计算。对于更深层的神经网络，必须定期应用重置操作以防止解密失败。然而，这个过程计算成本很高，由于涉及大量旋转操作，导致高延迟。此外，由于重置操作的大小很大，重置操作具有显著的内存占用。虽然新加密的密文从 L 个级别开始，但重置操作将可用级别减少到 $L-K$，因为 K 个级别在用于重置操作的多项式评估过程中被消耗。

计算复杂性。自举比旋转或密文-密文乘法慢两个数量级 [25]。密文-密文乘法比密文-明文乘法或加法慢两个数量级 [25]。

2.2. 同态CNN（HCNN）

同态卷积神经网络（HCNNs）是与HE支持在公式2中操作的运算兼容的卷积神经网络。我们根据是否使用自举，将HCNNs分为FHENets和LHENets。

LHENets包括CryptoNets [18], LoLa [5], Faster CryptoNets [11], ，而最近的FHENets包括MPCNN [28]和AutoFHE [1]。FHENets在图像分类数据集上实现了最先进的预测精度，但比LHENets具有更高的延迟。现有的HCNNs为了加速加密图像上的推理，主要集中在两个方面：卷积打包和多项式激活。

卷积打包是指有效地打包三维张量，以降低HE乘法和旋转的计算复杂度，用于卷积层 [24, 28]。MPCNN通过整合（1）重复打包和（2）复用打包来设计多路复用卷积 [28]。（1）一个循环多项式次数为 N 的密文可以打包 $\frac{N}{2}$ 个数字。给定一个具有 $x\in\mathbb{R}^d$ 的向量 $d\preceq\frac{N}{2}$，一个重复向量是 $x^{(M)}=[x,x,\cdots,x]^T$，包含 $M=\lfloor\frac{N}{2d}\rfloor$个 x 的副本。 $x^{(M)}$ 被加密以填充所有插槽。重复打包可以加速卷积和自举操作。更大的 M 会导致更快的推理。（2）当卷积步长超过1时，有效值之间的间隙会被引入，导致一些密文插槽未被使用。MPCNN通过将来自不同通道的数字打包到交替插槽中来解决此问题，有效地填充这些间隙并充分利用所有密文插槽，从而防止稀疏性。此外，通道是并行计算的，以加快卷积层。

多项式激活。HCNNs不能使用ReLU，因为它不是同态。因此，它们采用多项式激活（例如，单项式、切比雪夫多项式或埃尔米特多项式）来替代ReLU。单项式多项式被广泛使用，因为它们允许HCNNs被构建为传统的多项式网络。例如包括Cryp- toNets [18], LoLa [5], 和Faster CryptoNets [11]。然而，使用单项式多项式训练时，由于梯度爆炸，往往变得不稳定。使用切比雪夫多项式的极大极小近似可以实现ReLU函数的高精度近似 [27–29], ，但它们的高多项式度会导致过大的乘法深度。AESPA [39]通过引入低度埃尔米特多项式来减少乘法深度，并提出了逐基归一化来稳定训练。此外，基于搜索的Aut- oFHE [1] 探索逐层的混合度多项式，以进一步降低乘法深度。

CryptoFaceNet 模块。我们基于上述提到的先前进展，用于加速 HCNN 推理，具体采用了多路复用卷积 [28] 和低阶、基于基的埃尔米特激活 [39]。然而，我们的主要关注点在于解决端到端加密 FR 的其他未解决的挑战，包括处理高分辨率加密图像和最小化乘法深度以减少计算成本高昂的引导操作。我们通过CryptoFaceNet——一种新颖的架构设计——克服了这些挑战。

3. End-to-End Encrypted FR System

3.1. CryptoFace

FR Models [13, 26, 44] take advantage of the outstanding representation learning ability of CNNs [20] to extract discriminative features. Given a face image $x \in \mathbb{R}^{C\times H\times W}$, a neural network $f_\omega(\cdot)$ with trainable parameters ω, we have feature $y = f_\omega(x) \in \mathbb{R}^d$. To verify if two face images x_1, x_2 belong to the same individual, we compare their features $y_1 = f_\omega(x_1)$ and $y_2 = f_\omega(x_2)$ to obtain their similarity, *i.e.* $\text{Score}(y_1, y_2) = \|\frac{y_1}{\|y_1\|} - \frac{y_1}{\|y_1\|}\|^2 = 2 - 2\frac{y_1 y_2}{\|y_1\|\|y_2\|}$. If $\text{Score}(y_1, y_2)$ is smaller than a predefined threshold T, the two face images are classified as corresponding to the same person. We formulate this process as a Match function:

$$\text{Match}(y_1, y_2) = \text{Score}(y_1, y_2) - \text{T} \quad (3)$$

A neural network $f_\omega(\cdot)$ for FR is trained on a dataset $\mathcal{X}$ with M identities, each consisting of multiple face images. To train, we need to learn the feature center $W \in \mathbb{R}^{M\times d}$ with M d-dimensional feature centers, where d is the predefined feature dimension. We use ArcFace's [13] additive angular margin loss which is a modified cross-entropy loss:

$$\mathcal{L}_{\text{ArcFace}}(\omega) = -\log \frac{e^{s\cos(\theta_i+m)}}{e^{s\cos(\theta_i+m)} + \sum_{j=1, j\neq i}^{M} e^{s\cos\theta_j}} \quad (4)$$

Given a training sample x with identity i, the feature $y = f_\omega(x)$. The cosine similarity values between y and M d-dimensional features of the feature center are obtained. In Equation 4, θ_i is the angle between y and $M[i]$, while θ_j is the angle between y and $M[j], j \neq i$. The additive margin m is added to the angle θ_i. The scalar s increases the capacity of the unit ball.

Face verification involves two phases, *enrollment* and *verification*. In the enrollment stage, the client sends the server a reference face image and corresponding identity. The server employs the trained network to extract the feature from the reference face image and stores the feature and identity. In the verification stage, the client sends a probe face image and a claimed identity to the server. The server employs the same network to extract the feature from the probe face image and compares it with the reference feature indexed by the claimed identity.

Encrypted FR comprises two similar phases in our paper, *offline* and *online* as shown in Figure 2. The offline stage is similar to enrollment, and the online stage is analogous to verification. In the offline stage, the client generates a public key to encrypt the reference face image and sends the encrypted reference face image and the corresponding identity to the server. The server extracts the encrypted feature $[y_1] = f_\omega([x_1])$. The client does not need to wait for the offline stage to complete. However, during the online stage, the client must wait for the inference result; thus, the

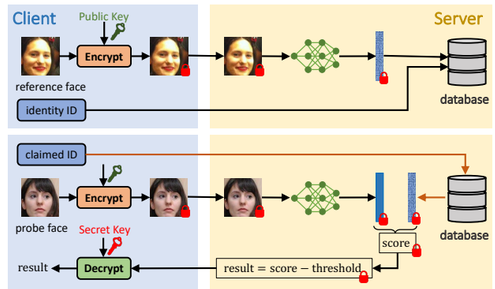


Figure 2. CryptoFace. Top: offline; bottom: online.

latency of the online stage is critical for real-world applications. In the online stage, the client encrypts a probe image $[x_2]$ and sends the encrypted probe face image and a claimed identity to the server. The server extracts the encrypted feature $[y_2] = f_\omega([x_2])$. Then, the server computes the match function (Equation 3) over the encrypted features, i.e., $\text{Match}([y_1], [y_2])$, and finally returns the resulting encrypted match result to the client. The client uses the secret key to decrypt the result and check if it is negative or positive to determine a match or no match, respectively.

Threat Model. Following the threat model used by existing encrypted FR systems [3, 15] and recent FHENets [1, 28], we assume two semi-honest parties: a client and a server. Under this model, at most one of these parties may be corrupted by an adversary [36]. Although both parties follow the agreed-upon protocol honestly, they may attempt to extract additional information by analyzing the data received from each other [36].

- If *the client is adversarial*, it may attempt to infer encrypted features stored on the server. However, since encrypted feature extraction is performed entirely on the server side without releasing intermediate features, the client receives only the matching result—a positive or negative scalar. Consequently, an adversarial client cannot infer the encrypted features stored on the server.
- If *the server is adversarial*, it may attempt to collect biometric data from the client. However, since the server only holds encrypted face images and features without access to the client's secret key, it cannot decrypt or infer any face images or features provided by the client.

3.2. CryptoFaceNet

Inspired by patch-based neural networks [14], CryptoFaceNet applies a mixture of PCNNs to extract local features and fuse these local features to obtain a global feature. Such a design significantly reduces FHE latency due to shallow PCNNs with lower multiplicative depth and parallelized evaluation under FHE.

Training process on cleartext data is shown in Figure 3

3. 端到端加密 FR 系统

3.1. CryptoFace

FR 模型 [13, 26, 44] 利用 CNNs [20] 卓越的表示学习能力来提取判别性特征。给定一张人脸图像 $x \in \mathbb{R}^{C\times H\times W}$，一个具有可训练参数 ω 的神经网络 $f_\omega(\cdot)$ 产生特征 $y = f_\omega(x) \in \mathbb{R}^d$。为了验证两张人脸图像 x_1、x_2 是否属于同一个人，我们比较它们的特征 $y_1 = f_\omega(x_1)$ 和 $y_2 = f_\omega(x_2)$ 以获得它们的相似度，即 $\text{Score}(y_1, y_2) = \|\frac{y_1}{\|y_1\|} - \frac{y_1}{\|y_1\|}\|^2 = 2 - 2\frac{y_1 y_2}{\|y_1\|\|y_2\|}$。如果 $\text{Score}(y_1, y_2)$ 小于预定义的阈值 T，这两张人脸图像被分类为对应于同一个人。我们将此过程表述为一个匹配函数：

$$\text{Match}(y_1, y_2) = \text{Score}(y_1, y_2) - \text{T} \quad (3)$$

用于FR的神经网络 $f_\omega(\cdot)$在包含M 身份的数据集 $\mathcal{X}$上训练，每个身份由多张人脸图像组成。为了训练，我们需要学习具有M d-维特征中心的特征中心 $W \in \mathbb{R}^{M\times d}$，其中 d 是预定义的特征维度。我们使用ArcFace的 [13] 加性角度边界损失，它是一种改进的交叉熵损失：

$$\mathcal{L}_{\text{ArcFace}}(\omega) = -\log \frac{e^{s\cos(\theta_i+m)}}{e^{s\cos(\theta_i+m)} + \sum_{j=1, j\neq i}^{M} e^{s\cos\theta_j}} \quad (4)$$

给定一个具有身份 i的训练样本 x，得到特征 $y = f_\omega(x)$。获得特征中心 y 与 M d-维特征之间的余弦相似度值。在公式4中，θ_i 是 y 与 $M[i]$之间的角度，而 θ_j是 y与 $M[j]$ $j \neq i$.之间的角度，加性边界 m 被加到角度 θ_i上。标量 s 增加了单位球的容量。

人脸验证涉及两个阶段，注册和验证。在注册阶段，客户端向服务器发送参考人脸图像和相应的身份。服务器使用训练好的网络从参考人脸图像中提取特征，并将特征和身份存储起来。在验证阶段，客户端向服务器发送探测人脸图像和声称的身份。服务器使用相同的网络从探测人脸图像中提取特征，并将其与通过声称身份索引的参考特征进行比较。

加密的 FR 在我们的论文中包含两个相似的阶段，离线和在线，如图 2 所示。离线阶段类似于注册，在线阶段类似于验证。在离线阶段，客户端生成一个公钥来加密参考人脸图像，并将加密的参考人脸图像和相应的身份信息发送到服务器。服务器提取加密的特征 $[y_1] = f_\omega([x_1])$。客户端不需要等待离线阶段完成。然而，在在线阶段，客户端必须等待推理结果；因此，

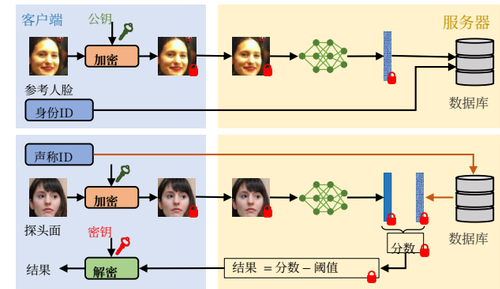


图2。CryptoFace。顶部：离线；底部：在线。

在线阶段的延迟对于实际应用至关重要。在在线阶段，客户端加密一个探测图像 $[x_2]$，并将加密的探测人脸图像和一个声称的身份信息发送到服务器。服务器提取加密的特征 $[y_2] = f_\omega([x_2])$。然后，服务器在加密的特征上计算匹配函数（公式 3），即 $\text{Match}([y_1], [y_2])$，并最终将加密的匹配结果返回给客户端。客户端使用密钥解密结果，并检查它是负值还是

分别用于确定匹配或未匹配。

威胁模型。遵循现有加密FR系统和近期FHENets所使用的威胁模型 [3, 15]，我们假设有两个半诚实方：一个客户端和一个服务器。在此模型下，最多只有一个方可能被对手 [36]破坏。尽管双方都诚实地遵循约定的协议，但它们可能会通过分析从对方接收到的数据 [36]来尝试提取额外信息。

- 如果客户端是恶意的，它可能会尝试推断服务器上存储的加密特征。然而，由于加密特征提取完全在服务器端执行且不释放中间特征，客户端仅收到匹配结果——一个正数或负数。因此，恶意的客户端无法推断服务器上存储的加密特征。

- 如果服务器是恶意的，它可能会尝试从客户端收集生物特征数据。然而，由于服务器仅持有加密的人脸图像和特征且无法访问客户端的密钥，它无法解密或推断客户端提供的任何人脸图像或特征。

3.2. CryptoFaceNet

受基于补丁的神经网络的启发 [14]，Crypto- Face Net 采用 PCNNs 的组合来提取局部特征，并将这些局部特征融合以获得全局特征。这种设计由于采用了具有较低乘法深度且在 FHE 下并行评估的浅层 PCNNs，显著降低了 FHE 延迟。

明文数据上的训练过程如图3所示

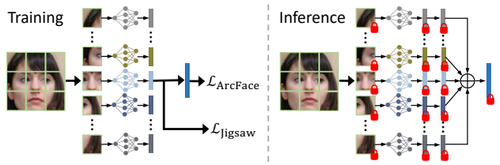


Figure 3. Left: cleartext training; right: encrypted inference.

(left). A two-dimensional face image $x \in \mathbb{R}^{C\times H\times W}$ is divided into a sequence of patches $x_i \in \mathbb{R}^{C\times P\times P} i = 1^L$, where $L = HW/P^2$. The image splitting approach follows that of Vision Transformer (ViT) [14]. A mixture of PCNNs, denoted as $\{f_{\omega_i}\}_{i=1}^L$, independently processes each patch x_i to generate local features $y_i \in \mathbb{R}^d$ for $1 \le i \le L$. These L local features are subsequently fused to form the global feature utilized for FR as follows,

$$y = y'A^T + b, \text{ where } y' = [y_1, y_2, \cdots, y_L] \quad (5)$$

where $A \in \mathbb{R}^{d\times dL}$ and $b \in \mathbb{R}^d$. The patch size is significantly smaller than the original image dimensions, i.e., $P \ll H$ or $P \ll W$, resulting in a reduced receptive field. This allows us to employ shallower PCNNs with lower multiplicative depth for each patch. Typically, each patch covers a small, distinctive facial structure, and the mixture of PCNNs learns separate filters for each patch instead of using weight-sharing [38]. The feature fusion process (Equation 5) encourages the PCNN mixture to capture inter-patch relationships effectively.

To train the mixture of PCNNs, we apply the ArcFace loss defined in Equation 4. Additionally, we introduce a jigsaw puzzle auxiliary task [6, 38, 40] to supplement positional information. This auxiliary task, previously shown to effectively capture positional details [6, 38, 40], naturally complements our mixture of PCNNs. Specifically, we use the local features $y_1, y_2, \cdots, y_L$ to predict their original positions (1, 2, $\cdots$, L) via a fully connected layer. Thus, the learned local features inherently encode positional information. The training objective of the mixture of PCNNs is:

$$\mathcal{L}(\omega, W, A, b) = \mathcal{L}_{\text{ArcFace}}(\omega, W, A, b) + \alpha\mathcal{L}_{\text{Jigsaw}}(\omega) \quad (6)$$

where $\omega = \{\omega_i\}_{i=1}^L$ are the parameters of all PCNNs, W is the feature center, A and b are feature fusion parameters, and α is the strength of jigsaw loss.

Inference. As illustrated in Figure 3 (right), the acceleration achieved by evaluating an encrypted face using the proposed mixture of PCNNs is two-fold. First, the multiplicative depth is significantly reduced, requiring only one bootstrapping operation per PCNN. Second, the evaluation of these L PCNNs can be performed in parallel, simplifying engineering implementation. However, the fusion step involves a large vector-matrix product $y'A^T$ as in Equation 5, which requires numerous computationally expensive homomorphic rotations. The mapping matrix $A \in \mathbb{R}^{d\times dL}$ is rectangular, making it incompatible with efficient HE vector-matrix multiplication approaches designed specifically for square matrices [19]. To address this issue, we rewrite A as $A = [A_1, A_2, \cdots, A_L]$, where each $A_i \in \mathbb{R}^{d\times d}$ for $1 \le i \le L$. Consequently, the vector-matrix multiplication can be decomposed as follows:

$$y = \sum_{i=1}^{L}\left(y_i A_i^T + \frac{b}{L}\right), \text{ where } y_i \in \mathbb{R}^d, A_i \in \mathbb{R}^{d\times d} \quad (7)$$

where the vector-matrix product $y_i A_i^T + b/L$ can be evaluated in parallel. Under FHE, the fusion function reduces to a simple *addition*, the computationally least expensive operation in FHE.

Scalability. Existing HCNNs described in Section 2.2 do not scale effectively to high-resolution face images. Increasing the size of tensors requires enlarging the degree (N) of the residue cyclotomic polynomial ring (see Section 2.1), leading to a substantial accumulation of latency. This occurs because ciphertext multiplication and rotation complexity under FHE scales as $\mathcal{O}(\ell^2 N \log N)$ [31]. In contrast, the proposed mixture of PCNNs is highly scalable to higher resolutions. By increasing the number of PCNNs, our approach achieves near-resolution-independent inference speed, due to the novel and efficient parallel evaluation strategy we introduce.

3.3. Homomorphic Architecture

Convolutional Block. As discussed in Section 2.2, we adopt FHE convolution from MPCNN [28] and the Hermite polynomial activation HerPN introduced by AESPA [39]. Figure 4 shows AESPA block and its FHE implementation provided by [1]. The AESPA block depth is 8 since one Conv consumes 2 levels, and one HerPN consumes 2 levels. We propose a depth-optimal *shifted* AESPA block. Figure 4 shows CryptoFaceNet blocks for stride $= 1$ and stride $= 2$. HerPN can be formulated as a degree-2 polynomial $ax^2 + bx + c$ with depth 2. We fuse the coefficient a to Conv weight and change the polynomial to $x^2 + \frac{b}{a}x + \frac{c}{a}$ with depth 1. Therefore, the proposed CryptoFaceNet block can save two levels.

Polynomial ℓ_2 Normalization. When computing the similarity score between two features y_1 and y_2, normalization of these features as $\frac{y_1}{\|y_1\|}$ and $\frac{y_2}{\|y_2\|}$ is necessary (Equation 3). ℓ_2 normalization can be expressed as $\frac{y}{\|y\|_2} = y \cdot \frac{1}{\sqrt{\sum_{i=1}^d y_i^2}}$. Under FHE, $[y_i]^2$ can be computed via ciphertext-ciphertext multiplication, and the summation $\sum_{i=1}^d [y_i]^2$ can be obtained using rotations. However, the primary challenge lies in approximating the non-linear function $q(t) = \frac{1}{\sqrt{t}}$. This is difficult as the domain of $q(t)$ is typically very wide (since

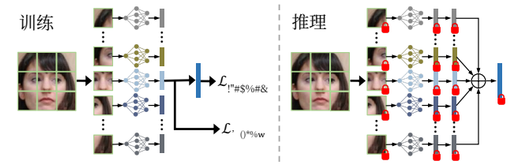


图3。左：明文训练；右：加密推理。

(左图)。二维人脸图像 $x \in \mathbb{R}^{C\times H\times W}$ 被分割成一系列块 $x_i \in \mathbb{R}^{C\times P\times P} i = 1^L$，其中 $L = HW/P^2$。图像分割方法遵循视觉Transformer（ViT） [14]的方法。PC-NNs的混合 $\{f_{\omega_i}\}_{i=1}^L$独立地处理每个块 x_i 以生成用于 $1 \le i \le L$的局部特征 $y_i \in \mathbb{R}^d$ 。这些 L 局部特征随后被融合形成用于FR的全局特征，具体如下，

$$y = y'A^T + b, \text{ where } y' = [y_1, y_2, \cdots, y_L] \quad (5)$$

其中 $A \in \mathbb{R}^{d\times dL}$ 和 $b \in \mathbb{R}^d$。补丁大小明显小于原始图像尺寸，即$P \ll H$ 或 $P \ll W$，导致感受野减小。这使得我们能够采用更浅的PCNNs，每个补丁的乘性深度更低。通常，每个补丁覆盖一个小而独特的面部结构，PCNN混合体为每个补丁学习独立的滤波器，而不是使用权重共享 [38]。特征融合过程（公式5）鼓励PCNN混合体有效捕获补丁间关系。

为了训练 PCNN 混合模型，我们应用了方程 4 中定义的 ArcFace 损失。此外，我们引入了一个拼图辅助任务 [6, 38, 40] 来补充位置信息。这个辅助任务之前已被证明能有效捕捉位置细节 [6, 38, 40], ，自然地补充了我们的 PCNN 混合模型。具体来说，我们使用局部特征 $y_1, y_2, \cdots, y_L$ 通过全连接层预测它们原始的位置 (1, 2, $\cdots$, L)。因此，学习到的局部特征本质上编码了位置信息。PCNN 混合模型的训练目标是：

$$\mathcal{L}(\omega, W, A, b) = \mathcal{L}_{\text{ArcFace}}(\omega, W, A, b) + \alpha\mathcal{L}_{\text{Jigsaw}}(\omega) \quad (6)$$

其中 $\omega = \{\omega_i\}_{i=1}^L$ 是所有 PCNN 的参数， W 是特征中心， A 和 b 是特征融合参数， α 是拼图损失的强度。

推理。如图3（右）所示，使用所提出的PCNN混合模型对加密人脸进行评估所实现的加速效果是双重的。首先，乘法深度显著降低，每个PCNN仅需一次引导操作。其次，这些 L PCNN的评估可以并行执行，简化了工程实现。然而，融合步骤涉及一个大型向量矩阵乘法 $y'A^T$，如方程5所示，

这需要大量计算成本高昂的同态旋转。映射矩阵 $A \in \mathbb{R}^{d\times dL}$ 是矩形的，使其与为方形矩阵 [19]专门设计的有效HE向量矩阵乘法方法不兼容。为解决此问题，我们将 A重写为 $A = [A_1, A_2, \cdots, A_L]$，其中每个 $A_i \in \mathbb{R}^{d\times d}$用于$1 \le i \le L$。因此，向量矩阵乘法可以分解如下：

$$y = \sum_{i=1}^{L}\left(y_i A_i^T + \frac{b}{L}\right), \text{ where } y_i \in \mathbb{R}^d, A_i \in \mathbb{R}^{d\times d} \quad (7)$$

其中向量矩阵乘法 $y_i A_i^T + b/L$ 可以并行计算。在 FHE 下，融合函数简化为简单的加法，这是 FHE 中计算成本最低的操作。

可扩展性。第2.2节中描述的现有HCNNs无法有效地扩展到高分辨率人脸图像。增加张量的大小需要扩大残差循环多项式环的次数(N)（参见第2.1节），导致延迟显著累积。这是因为FHE下的密文乘法和旋转复杂度按 $\mathcal{O}(\ell^2 N \log N)$ [31]的规模增长。相比之下，我们提出的PCNN混合模型在高分辨率上具有高度可扩展性。通过增加PCNN的数量，我们的方法实现了近乎分辨率无关的推理速度，这归因于我们引入的新颖且高效的并行评估策略。

3.3. 同态架构

卷积块。如第2.2节所述，我们采用来自MPCNN [28]的FHE卷积和AESPA引入的Hermite多项式激活函数HerPN。图4展示了AESPA块及其由 [1]提供的FHE实现。AESPA块的深度为8，因为一个Conv消耗2级，一个HerPN消耗2级。我们提出了一种深度优化的移位AESPA块。图4展示了CryptoFaceNet块，用于步长 $= 1$ 和步长 $= 2$。HerPN可以表示为次数为2的多项式 $ax^2 + bx + c$，深度为2。我们将系数 a与Conv权重融合，并将多项式更改为深度为1的 $x^2 + \frac{b}{a}x + \frac{c}{a}$。 - 因此，所提出的CryptoFaceNet块可以节省两个级别。

多项式 ℓ_2 归一化。在计算两个特征 y_1 和 y_2之间的相似度分数时，需要对这些特征进行 $\frac{y_1}{\|y_1\|}$ 和 $\frac{y_2}{\|y_2\|}$ 归一化（公式3）。ℓ_2 归一化可以表示为 $\frac{y}{\|y\|_2} = y \cdot \frac{1}{\sqrt{\sum_{i=1}^d y_i^2}}$。在FHE下，$[y_i]^2$ 可以通过密文-密文乘法计算，而求和 $\sum_{i=1}^d [y_i]^2$ 可以使用旋转获得。然而，主要挑战在于近似非线性函数 $q(t) = \frac{1}{\sqrt{t}}$。这很困难，因为 $q(t)$的域通常非常宽（因为

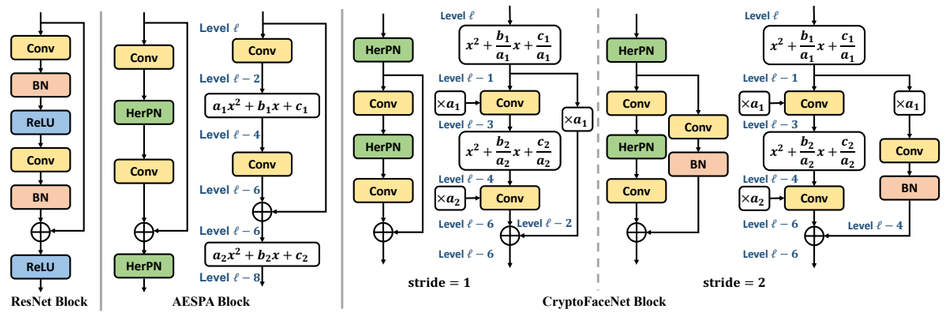


Figure 4. Convolutional blocks and their FHE implementations. Left: ResNet [20]; middle: AESPA [39]; right: CryptoFaceNet.

$t = |y|_2^2$). A Taylor expansion around $t = t_0$ would result in a polynomial with excessively high degree, and minimax approximation [27] is not directly applicable since $q(t)$ cannot be effectively scaled into the $[-1, 1]$ interval it requires. Observing that $q(t) = \frac{1}{\sqrt{t}}$ is a strictly decreasing function and we propose using a simpler polynomial approximation: $p(t) = \beta_2 t^2 + \beta_1 t + \beta_0$ to approximate $q(t)$. Unlike the minimax approximation that explicitly minimizes $|p(t) - q(t)|$, we instead ensure $p(t)$ has a similar shape to $q(t)$ within its relevant domain. We achieve this by selecting three control points t_1, t_2, and t_3, along with their corresponding reference values $\frac{1}{\sqrt{t_1}}$, $\frac{1}{\sqrt{t_2}}$, and $\frac{1}{\sqrt{t_3}}$, respectively. Solving these equations yields coefficients β_2, β_1, and β_0 that fit our polynomial to the distribution of data. Specifically, we set the control points based on the distribution of t as follows: $t_1 = \mathrm{Mean}(t) - \mathrm{Std}(t)$, $t_2 = \mathrm{Mean}(t)$, $t_3 = \mathrm{Mean}(t) + \mathrm{Std}(t)$. Thus, the resulting polynomial $p(t)$ is distribution-aware, and its coefficients (β_2, β_1, and β_0) are determined by the underlying data distribution—similar to the score thresholding procedure. The polynomial approximation has a multiplicative depth of only 2, ensuring computational efficiency. Consequently, we estimate the matching threshold (i.e., T in Equation 3) to accommodate the polynomial-based ℓ_2 normalization.

Adaptive Average Pooling. MPCNN [28] originally employs adaptive average pooling with an output size of $(1, 1)$. To better preserve structural information crucial for FR, we customized it to have an output size of $(2, 2)$, resulting in 256-dimensional features. Our $(2, 2)$ pooling is implemented as four separate $(1, 1)$ pooling operations, which permute elements of the feature vector. So, we also rearranged the fusion matrix A accordingly (Equation 5) in the cleartext domain.

CryptoFaceNet Architecture is shown in Figure 5. To reduce depth consumption, CryptoFaceNet fuses the Linear and BatchNorm1D layers to a single Linear. The aggregation of features is a simple ciphertext addition. CryptoFaceNet only uses one bootstrapping operation.

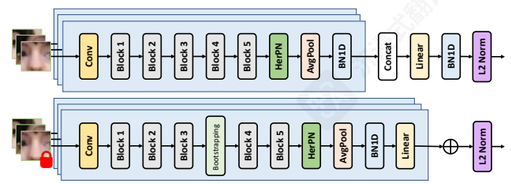


Figure 5. CryptoFaceNet. Top: training; bottom: inference.

4. Experiments

Datasets. (1) Training dataset: we use WebFace4M, a subset of WebFace260M [48] to train CryptoFace and baselines on cleartext data in Pytorch. WebFace4M has 205,990 identities and 4,235,242 face images in total. (2) Test datasets: we follow AdaFace [26] to benchmark CryptoFace and baselines on five standard test datasets [22]: LFW [21], AgeDB [37], CALFW [47], CPLFW [46], CFP-FP [42]. The image resolution is 112×112. In our experiments, we resize face images to small (64×64), medium (96×96), and high-resolution (128×128), which satisfies use cases corresponding to edge FR, embedded FR, and cloud FR in the wild. CFP-FP has 7,000 pairs, resulting in 14,000 face images, while others have 6,000 pairs, resulting in 12,000 face images. We report the one-to-one verification accuracy on the encrypted test datasets.

FHE Library and Hardware. We follow MPCNN [28] and AutoFHE [1] to adopt SEAL [41] library and report latency and RAM footprint on Amazon AWS r5.24xlarge. The modified SEAL [28] incorporates bootstrapping.

Baselines. As there is no prior end-to-end encrypted FR, we adopt two FHENets MPCNN and AutoFHE (see Section 2.2) as our baselines because they report the SoTA performance on CIFAR image classification. Our C++ implementation of CryptoFace is built on top of MPCNN and AutoFHE. To take 64×64 face images as input, we change the stride of the very first convolutional layer from 1 to 2 to ensure that a single ciphertext can pack any intermediate tensors. The new output layers used for FR are

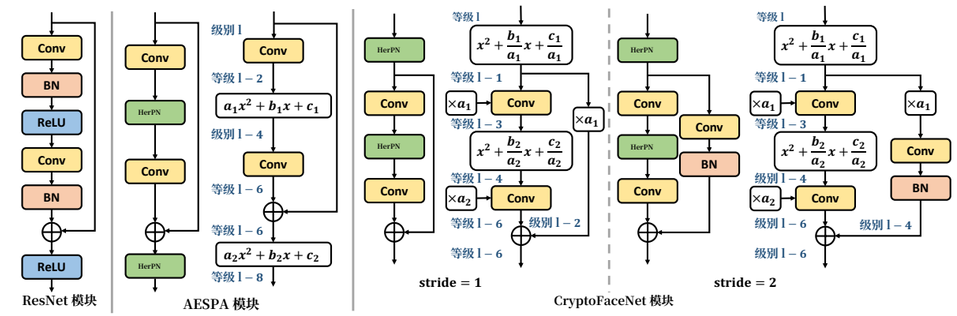


图 4。卷积块及其 FHE 实现。左：ResNet [20]；中：AESPA [39]；右：CryptoFaceNet。

$t = |y|_2^2$)。在 $t = t_0$ 周围的泰勒展开会产生一个度数过高的多项式，而极小极大近似 [27] 不直接适用，因为 $q(t)$不能有效地缩放到它所需的 $[-1, 1]$ 区间。观察到 $q(t) = \frac{1}{\sqrt{t}}$ 是一个严格递减的函数，我们提出使用一个更简单的多项式近似： $p(t) = \beta_2 t^2 + \beta_1 t + \beta_0$ 来近似 $q(t)$。与显式最小化$|p(t) - q(t)|$的极小极大近似不同，我们确保 $p(t)$在其相关域内与 $q(t)$具有相似的形状。我们通过选择三个控制点 t_1、 t_2和 t_3，以及它们对应的参考值 $\frac{1}{\sqrt{t_1}}$、 $\frac{1}{\sqrt{t_2}}$和 $\frac{1}{\sqrt{t_3}}$来实现这一点。求解这些方程得到系数 β_2、 β_1和 β_0，使我们的多项式拟合数据的分布。具体来说，我们根据 t 的分布设置控制点如下： $t_1 = \mathrm{Mean}(t) - \mathrm{Std}(t)$， $t_2 = \mathrm{Mean}(t)$ $t_3 = \mathrm{Mean}(t) + \mathrm{Std}(t)$。因此，得到的多项式 $p(t)$是分布感知的，其系数(β_2、 β_1和 β_0)由底层数据分布决定——类似于分数阈值化过程。多项式近似只有2的乘法深度，确保了计算效率。因此，我们估计匹配阈值（即公式3中的 T）以适应基于多项式的 ℓ_2 归一化。

自适应平均池化。MPCNN [28]最初采用自适应平均池化，输出大小为(1, 1)。为了更好地保留对FR至关重要的结构信息，我们将它定制为输出大小为(2, 2)，从而得到256维特征。我们的(2,2)池化实现为四个独立的(1, 1)池化操作，这些操作会置换特征向量的元素。因此，在明文域中，我们也相应地重新排列了融合矩阵 A （公式5）。

CryptoFaceNet 架构如图 5 所示。为减少深度消耗，CryptoFaceNet 将 Linear 和 BatchNorm1D 层融合为单个 Linear。特征聚合采用简单的密文加法。CryptoFaceNet 仅使用一次自举操作。

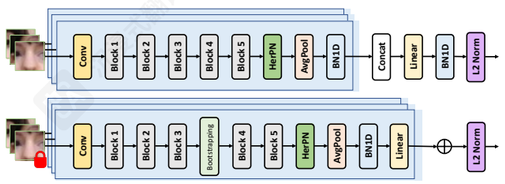


图 5. CryptoFaceNet。顶部：训练；底部：推理。

4. 实验

数据集。(1) 训练数据集：我们使用 WebFace4M（WebFace260M 的子集）[48] 在 Pytorch 中使用明文数据训练 CryptoFace 和基线模型。WebFace4M 共有 205,990 个身份标识和 4,235,242 张人脸图像。(2) 测试数据集：我们遵循 AdaFace [26]，在五个标准测试数据集 [22] 上对 CryptoFace 和基线模型进行基准测试：LFW [21],AgeDB [37], CALFW [47], CPLFW [46], CFP-FP [42]。图像分辨率为 112×112。在我们的实验中，我们将人脸图像调整为小 (64×64)、中 (96×96) 和高分辨率 (128×128)，这满足了野外场景中边缘 FR、嵌入式 FR 和云 FR 对应的使用场景。CFP-FP 有 7,000 对，产生 14,000 张人脸图像，而其他数据集有 6,000 对，产生 12,000 张人脸图像。我们在加密测试数据集上报告了一对一验证准确率。

FHE 库和硬件。我们遵循 MPCNN [28]和 AutoFHE [1] 来采用 SEAL [41] 库，并在 Amazon AWS r5.24xlarge 上报告延迟和 RAM 脚印。修改后的 SEAL [28] 集成了自举功能。

基准线。由于没有先前的端到端加密 FR，我们采用两个 FHENets MPCNN 和 AutoFHE（见第 2.2 节）作为我们的基准线，因为它们报告了在 CIFAR 图像分类上的 SoTA 性能。我们的 C++ ryptoFace 实现基于 MPCNN 和 AutoFHE。为了将 64×64 人脸图像作为输入，我们将第一个卷积层的步长从 1 改为 2，以确保单个密文可以打包任何中间张量。用于 FR 的新输出层是

Method	Backbone				Dataset						Latency(s)	RAM
	Network	Params	#Boot	Res	LFW	AgeDB	CALFW	CPLFW	CFP-FP	Avg		
MPCNN [28]	ResNet32	0.53M	31	64	97.02	83.02	87.00	78.90	82.07	85.60	7,367	286G
	ResNet44	0.72M	43	64	98.27	87.45	90.85	83.72	87.90	89.64	9,845	286G
AutoFHE [1]	ResNet32	0.53M	8	64	93.53	80.88	85.40	75.67	77.96	82.69	4,001	286G
CryptoFace	CryptoFaceNet4	0.94M	1	64	98.87	89.45	91.60	81.98	85.21	89.42	1,364	269G
	CryptoFaceNet9	2.12M		96	99.18	91.38	93.32	84.23	86.81	90.99	1,395	276G
	CryptoFaceNet16	3.78M		128	98.78	92.90	93.73	83.95	87.94	91.46	1,446	277G

Table 1. Experiments on end-to-end FR on FHE.

Method	Backbone	Conv	BN	Residual	AvgPool	Linear	Activation	L2 Norm	Match	Bootstrapping	Other
MPCNN [28]	ResNet32	896s (12.17%)	28s (0.38%)	0.5s (0.01%)	46s (0.62%)	807s (10.96%)	583s (7.92%)	5s (0.07%)	2s (0.02%)	4991s (67.76%)	7s (0.09%)
	ResNet44	1214s (12.33%)	39s (0.39%)	0.7s (0.01%)	46s (0.46%)	807s (8.19%)	807s (8.20%)	5s (0.05%)	2s (0.02%)	6917s (70.27%)	7s (0.08%)
AutoFHE [1]	ResNet32	1966s (49.14%)	28s (0.69%)	1.7s (0.04%)	38s (0.95%)	658s (16.43%)	17s (0.43%)	4s (0.10%)	1s (0.03%)	1274s (31.84%)	14s (0.34%)
CryptoFace	CryptoFaceNet4	858s (62.93%)	2s (0.16%)	0.1s (0.01%)	26s (1.94%)	277s (20.30%)	13s (0.94%)	2s (0.11%)	0.3s (0.02%)	141s (10.34%)	44s (3.26%)

Table 2. Latency of operations on FHE for 64×64 encrypted face images.

AdaptiveAvgPool2d((2, 2)) $\mapsto$ Linear(256, 256) $\mapsto$ BatchNorm1d(256). AutoFHE is a search-based approach and reports search results on the CIFAR dataset. Extending the search algorithm from a small CIFAR dataset to the much larger WebFace dataset is challenging, so we use the search result on CIFAR and transfer it to the WebFace dataset.

Parameters. (1) Training: we adopt the parameters from AdaFace [26] without tuning. We set the learning rate to 0.05, epochs to 26, batch size to 256, momentum to 0.9, and weight decay to 0.0005. We use the SGD optimizer with multi-step scheduler. The learning rate is scaled by $0.1\times$ at epochs 12, 20, and 24. When we train CryptoFace and AutoFHE, we clip the gradient to 1 as suggested by AutoFHE. The patch size is set to 32×32. We set the strength (α) of the jigsaw loss to 0.005. (2) FHE: To meet 128-bit security [10], we use the same CKKS parameters as MPCNN and AutoFHE. The degree of the cyclotomic ring is 2^{16} and Hamming weight is 192. We set the default modulus to 46 bits and the special modulus to 51 bits [1, 28].

4.1. End-to-End Encrypted FR on FHE

We follow the standard 10-fold cross-validation [13, 26] to benchmark CryptoFace and baselines on the encrypted face datasets. For each split, nine groups (cleartext) are used to estimate the threshold and polynomial approximation of ℓ_2 normalization, while the standalone group (ciphertext) is used to test performance. We report the average verification accuracy of 10-fold cross-validation for each dataset as shown in Table 1. For a pair of face images, we consider the first as the reference and the second as the probe. Each experiment includes an offline and an online stage (see Section 3.1). In Table 1, we report the online latency. We use numbers of patches (*i.e.* 4, 9, and 16) to denote CryptoFaceNet for different resolutions 64×64, 96×96, and 128×128, respectively.

For encrypted face images at resolution 64×64, MPCNN with ResNet44 shows the highest accuracy (89.64%) but a prohibitively large latency (9,845 seconds). CryptoFace greatly accelerates encrypted face recognition by $\mathbf{7.2}\times$ which translates to savings of 8,481 seconds. We only observe a negligible accuracy drop of 0.22%. MPCNN with ResNet32 is a faster version but only achieves 85.60% accuracy. CryptoFace speeds up inference by $\mathbf{5.4}\times$ and increases FR performance by $+\mathbf{3.82}\%$. AutoFHE is much faster than MPCNN. However, the transferred AutoFHE achieves 82.64% on encrypted FR with a latency of 4,001 seconds. Compared to AutoFHE, CryptoFace accelerates inference by $\mathbf{2.9}\times$ and improves the encrypted FR performance by $+\mathbf{6.73}\%$. CryptoFace also reduces RAM footprint by **17G** since we only need one bootstrapping operation, while the baselines require three operations for different repeated packing copies M (see Section 2.2). The experimental results demonstrate the effectiveness of CryptoFace, the proposed end-to-end encrypted FR. The proposed CryptoFaceNet is an efficient FHENet thanks to the mixture of PCNNs with simple, yet effective, parallelization.

We also analyze how the resolution of encrypted face images impacts verification accuracy, latency, and RAM footprint, as shown in Table 1. When the resolution increases from 64×64 to 96×96 and 128×128, CryptoFace can take advantage of high-resolution face images to effectively improve FR performance. Compared to CryptoFaceNet4, CryptoFaceNet9 and CryptoFaceNet16 increase the accuracy by $+\mathbf{1.57}\%$ and $+\mathbf{2.04}\%$, respectively. CryptoFace can maintain a nearly constant latency (1,364 to 1,446 seconds) even as the image resolution increases from 64×64 to 128×128. CryptoFace slightly increases the RAM footprint from 269G to 277G. The results demonstrate that CryptoFace is scalable to high-resolution images and can satisfy different requirements in real-world secure FR applications.

4.2. Operation Latency

Table 2 lists detailed latency of different operations. Bootstrapping operations dominate the latency of MPCNN, around 70%. AutoFHE successfully removes most bootstrapping operations by using mixed-degree polynomial activations. CryptoFace fundamentally decreases the depth of networks and significantly reduces the number of bootstrap-

方法	骨干				数据集						延迟(s)	RAM
	网络	参数	#启动	Res	LFW	AgeDB	CALFW	CPLFW	CFP-FP	Avg		
MPCNN [28]	ResNet32	0.53M	31	64	97.02	83.02	87.00	78.90	82.07	85.60	7,367	286G
	ResNet44	0.72M	43	64	98.27	87.45	90.85	83.72	87.90	89.64	9,845	286G
AutoFHE [1]	ResNet32	0.53M	8	64	93.53	80.88	85.40	75.67	77.96	82.69	4,001	286G
CryptoFace	CryptoFaceNet4	0.94M	1	64	98.87	89.45	91.60	81.98	85.21	89.42	1,364	269G
	CryptoFaceNet9	2.12M		96	99.18	91.38	93.32	84.23	86.81	90.99	1,395	276G
	CryptoFaceNet16	3.78M		128	98.78	92.90	93.73	83.95	87.94	91.46	1,446	277G

表1. FHE上的端到端FR实验

方法	骨干网络	Conv	BN	残差	平均池化	线性	激活	L2范数	匹配	自举	其他
MPCNN [28]	ResNet32	896s (12.17%)	28s (0.38%)	0.5s (0.01%)	46s (0.62%)	807s (10.96%)	583s (7.92%)	5s (0.07%)	2s (0.02%)	4991s (67.76%)	7s (0.09%)
	ResNet44	1214s (12.33%)	39s (0.39%)	0.7s (0.01%)	46s (0.46%)	807s (8.19%)	807s (8.20%)	5s (0.05%)	2s (0.02%)	6917s (70.27%)	7s (0.08%)
AutoFHE [1]	ResNet32	1966s (49.14%)	28s (0.69%)	1.7s (0.04%)	38s (0.95%)	658s (16.43%)	17秒 (0.43%)	4秒 (0.10%)	1秒 (0.03%)	1274秒 (31.84%)	14秒 (0.34%)
CryptoFace	CryptoFaceNet4	858s (62.93%)	2s (0.16%)	0.1s (0.01%)	26s (1.94%)	277s (20.30%)	13s (0.94%)	2s (0.11%)	0.3s (0.02%)	141s (10.34%)	44s (3.26%)

表2。 64×64 加密人脸图像上FHE操作的延迟。

AdaptiveAvgPool2d((2, 2)) $\mapsto$ Linear(256, 256) $\mapsto$ BatchNorm1d(256)。AutoFHE 是一种基于搜索的方法，并在 CIFAR 数据集上报告搜索结果。将搜索算法从小型 CIFAR 数据集扩展到远大的 WebFace 数据集是具有挑战性的，因此我们使用 CIFAR 上的搜索结果并将其迁移到 WebFace 数据集。

参数。 (1) 训练：我们采用来自 AdaFace [26] 的参数而不进行调优。我们将学习率设置为 0.05，训练轮次设置为 26，批大小设置为 256，动量为 0.9，权重衰减设置为 0.0005。我们使用带有多步调度的 SGD 优化器。学习率在训练轮次 12、20 和 24 时通过 $0.1\times$ 进行缩放。在训练 CryptoFace 和 AutoFHE 时，我们按照 AutoFHE 的建议将梯度裁剪到 1。我们设置补丁大小为 32×32。我们将拼图损失的强度 (α) 设置为 0.005。 (2) FHE：为了满足 128 位安全性 [10], ，我们使用与 MPCNN 和 AutoFHE 相同的 CKKS 参数。循环域的次数为 2^{16}，汉明重为 192。我们将默认模数设置为 46 位，特殊模数设置为 51 位 [1, 28]。

4.1. 基于FHE的端到端加密FR

我们遵循标准的10折交叉验证方法 [13, 26]，在加密人脸数据集上对CryptoFace和基线模型进行基准测试。对于每个划分，使用九个组（明文）来估计ℓ_2归一化的阈值和多项式近似，而独立的组（密文）用于测试性能。我们报告了每个数据集10折交叉验证的平均验证准确率，如表1所示。对于一对人脸图像，我们将第一个视为参考，第二个视为探测。每个实验包括离线和在线两个阶段（见第3.1节）。在表1中，我们报告了在线延迟。我们使用补丁数量（即4、9和16）来表示不同分辨率 64×64、 96×96和128×128的CryptoFaceNet。

Forencrypted 人脸图像在分辨率 64×64 下，MPCNN 与 ResNet44 显示出最高的准确率（89.64%），但延迟过高（9,845 秒）。CryptoFace 通过 $\mathbf{7.2}\times$ 大幅加速了加密人脸识别。这相当于节省了 8,481 秒。我们仅观察到 0.22% 的可忽略准确率下降。MPCNN 与 ResNet32 是更快版本，但仅达到 85.60% 的准确率。CryptoFace 通过 $\mathbf{5.4}\times$ 加速了推理，并通过 $+\mathbf{3.82}\%$ 提高了 FR 性能。AutoFHE 比 MPCNN 快得多。然而，传输的 AutoFHE 在加密 FR 上达到 82.64%，延迟为 4,001 秒。与 AutoFHE 相比，CryptoFace 通过 $\mathbf{2.9}\times$ 加速了推理，并通过 $+\mathbf{6.73}\%$ 提高了加密 FR 性能。CryptoFace 还将 RAM 脚印减少了 17G，因为我们只需要一次引导操作，而基线需要三次操作用于不同的重复打包副本 M （见第 2.2 节）。实验结果表明了 CryptoFace 的有效性，即所提出的端到端加密 FR。所提出的 CryptoFaceNet 是一个高效的 FHENet，这得益于将 PCNNs 与简单但有效的并行化相结合。

我们还分析了加密人脸图像的分辨率如何影响验证准确率、延迟和RAM占用，如表1所示。当分辨率从 64×64 增加到 96×96 和 128×128时，CryptoFace可以利用高分辨率人脸图像有效提升FR性能。与CryptoFaceNet4相比，CryptoFaceNet9和CryptoFaceNet16分别将准确率提高了 $+\mathbf{1.57}\%$和 $+\mathbf{2.04}\%$。即使图像分辨率从 64×64 增加到128×128，CryptoFace也能保持近乎恒定的延迟（1,364到1,446秒）。CryptoFace的RAM占用从269G略微增加到277G。结果表明，CryptoFace可扩展至高分辨率图像，并能满足现实世界安全FR应用中的不同需求。

4.2. 运行延迟

表2列出了不同操作的详细延迟。自举操作主导了MPCNN的延迟，约占70%。AutoFHE通过使用混合度多项式激活成功移除了大多数自举操作。CryptoFace从根本上减少了网络的深度，并显著减少了自举-

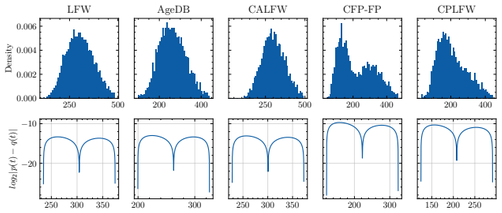


Figure 6. ℓ_2 polynomial approximation. Top: the distribution (domain) of $q(t) = 1/\sqrt{t}$, bottom: the approximation error $\log_2|p(t) - q(t)|$.

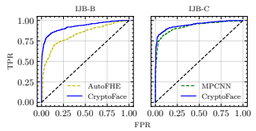
ping operations to *one* with bootstrapping only contributing 10% to the total latency. The parallelism of CryptoFaceNet on FHE introduces an acceptable overhead, around $< 3.26\%$ (Other). CryptoFace and MPCNN (ResNet32) spend roughly equal time evaluating convolutional layers. However, MPCNN (ResNet32) has 31 convolutional layers, and one patch CNN only has 13 (including two residual convolutional layers) since CryptoFaceNet convolutional layers take high-level ciphertext as input, leading to higher latency. Both AutoFHE and CryptoFace benefit from low-degree polynomial activations because expensive ciphertext-ciphertext multiplications are decreased. We apply the average pooling with the output size $(2, 2)$ to get 256-dimensional features (see Section 3.3). The $(2, 2)$ average pooling is slower than $(1, 1)$ averaging pooling used by the original versions of MPCNN and AutoFHE because it introduces more rotations and multiplications. However, FR necessitates high-dimensional features to preserve more structural information. The proposed polynomial approximation of ℓ_2 normalization and the feature matching are very efficient under FHE. CryptoFaceNet is a depth-optimized homomorphic neural architecture (see Section 3.3) demonstrating lower latency over different FHE operations.

4.3. Polynomial L2 Approximation

Figure 6 (top) shows the distribution of $\|y\|_2^2$, namely the domain of $q(t) = 1/\sqrt{t}$. We propose a distribution-aware polynomial approximation $p(t) \mapsto q(t)$ (see Section 3.3). Figure 6 (bottom) shows that the approximation error is $\log_2|p(t) - q(t)| \leq 2^{-10}$. Table 2 shows the latency of the polynomial ℓ_2 approximation is 0.3 to 2 seconds on FHE only consuming 0.02% of inference time. Thus, the polynomial ℓ_2 approximation is accurate and efficient.

4.4. Mixed-Quality FR Benchmarks

The five standard FR datasets used in our experiments are regarded as high-quality FR benchmarks. They can satisfy the most real-world secure FR applications. IJB-B [45] and IJB-C [33] FR datasets include mixed-quality face images and are used to test FR models for challenging FR tasks.



IJB-B	AutoFHE	**CryptoFace**
Time	4,001s	1,364s (↓ **2,637s**)
AUC	0.825	0.926 (↑ **0.1**)
IJB-C	**MPCNN**	**CryptoFace**
Time	7,367s	1,364s (↓ **6,003s**)
AUC	0.927	0.946 (↑ **0.02**)

Figure 7. Experiments on IJB-B and IJB-C.

Figure 7 shows experimental results on encrypted IJB-B and IJB-C on FHE. Due to the prohibitively high computational cost, we randomly sample 1,200 pairs (with 50% positive pairs and 50% negative pairs) from each dataset. We use ROC (Receiver Operating Characteristic) curve and AUC (Area Under the Curve) to qualify the FR performance of CryptoFace and baselines. From Figure 7, CryptoFace consistently shows better performance on IJB-B or IJB-C on FHE compared to AutoFHE and MPCNN.

4.5. Identification

In this paper, we primarily focus on the face verification task, as discussed in Section 3.1. However, CryptoFace can be readily extended to face identification scenarios, such as one-to-many face matching. In Table 3, we report experimental results for a $1:128$ closed-set rank retrieval task under FHE, using a randomly selected subset of LFW consisting of 128 pairs. The latency difference solely arises from the feature extraction stage, as previously reported. We evaluate performance using Rank-1 and Rank-5 accuracy as metrics. The experimental results confirm that CryptoFace is also effective for face identification tasks.

Method	Rank-1 Acc.	Rank-5 Acc.
MPCNN	88.28	97.66
CryptoFace	92.19 (↑ **3.91%**)	98.44 (↑ **0.78%**)

Table 3. 1:128 closed-set retrieval.

5. Conclusion

This paper introduced CryptoFace, the first end-to-end encrypted face recognition system using fully homomorphic encryption (FHE). Once face images are encrypted, all subsequent operations like feature extraction, matching, and score thresholding are performed in the encrypted domain without decryption. The key idea behind CryptoFace is CryptoFaceNet, a novel architecture which is a mixture of shallow patch convolutional neural networks optimized for FHE compatibility and mitigating the steep computational burden of encrypted inference. Experimental results on standard face recognition benchmarks show that CryptoFace is $7\times$ faster than SoTA FHENets while achieving better verification performance. CryptoFace can effectively process high-resolution encrypted face images to improve verification accuracy by $+2\%$ while maintaining near-resolution-independent latency. CryptoFace will facilitate the deployment of secure face recognition systems in applications requiring strict privacy and security guarantees.

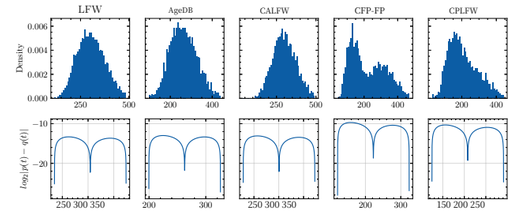


图6。 ℓ_2 多项式逼近。顶部： $q(t) = 1/\sqrt{t}$的分布（域），底部：逼近误差$\log_2|p(t) - q(t)|$。

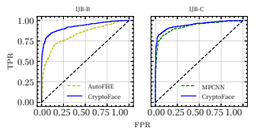
ping操作的延迟，其中只有自举操作占总延迟的10%。CryptoFaceNet在FHE上的并行性引入了可接受的开销，约为$< 3.26\%$（其他）。CryptoFace和MPCNN（ResNet32）在评估卷积层上花费的时间大致相等。然而，MPCNN（ResNet32）有31个卷积层，而一个补丁CNN只有13个（包括两个残差卷积层），因为CryptoFaceNet的卷积层以高级密文为输入，导致更高的延迟。AutoFHE和CryptoFace都受益于低度多项式激活，因为昂贵的密文-密文乘法被减少了。我们对输出大小为（2，2）的平均池化应用平均池化以获得256维特征（见第3.3节）。（2，2）平均池化比MPCNN和AutoFHE原始版本使用的（1，1）平均池化慢，因为它引入了更多的旋转和乘法。然而，FR需要高维特征以保留更多信息结构。所提出的 ℓ_2 归一化多项式近似和特征匹配在FHE下非常高效。CryptoFaceNet是一种深度优化的同态神经网络架构（见第3.3节），在不同FHE操作上表现出更低的延迟。

4.3. 多项式L2逼近

图6（顶部）显示了 $\|y\|_2^2$的分布，即$q(t) = 1/\sqrt{t}$. 的域。我们提出了一种分布感知的多项式逼近 $p(t) \mapsto q(t)$（见第3.3节）。图6（底部）显示逼近误差为$\log_2|p(t) - q(t)| \leq 2^{-10}$。表2显示多项式 ℓ_2 逼近在FHE上仅消耗0.02%的推理时间，延迟为0.3到2秒。因此，多项式 ℓ_2 逼近是准确且高效的。

4.4. 混合质量 FR 基准测试

我们实验中使用的五个标准 FR 数据集被视为高质量的 FR 基准。它们能满足大多数现实世界的安全 FR 应用。IJB-B [45] 和 IJB-C [33] FR 数据集包含混合质量的图像，用于测试 FR 模型在具有挑战性的 FR 任务中的性能。



IJB-B	AutoFHE	**CryptoFace**
Time	4,001s	1,364秒 (↓ **2,637秒**)
AUC	0.825	0.926 (↑ **0.1**)
IJB-C	**MPCNN**	**CryptoFace**
Time	7,367s	1,364秒 (↓ **6,003秒**)
AUC	0.927	0.946 (↑ **0.02**)

图7。IJB-B和IJB-C的实验。

图 7 展示了在 FHE 上对加密 IJB-B 和 IJB-C 的实验结果。由于计算成本过高，我们从每个数据集中随机采样 1,200 对（其中 50% 为正对，50% 为负对）。我们使用 ROC（接收者操作特征曲线）和 AUC（曲线下面积）来评估 CryptoFace 和基线的 FR 性能。从图 7 可以看出，与 AutoFHE 和 MPCNN 相比，CryptoFace 在 FHE 上的 IJB-B 或 IJB-C 上始终表现出更好的性能。

4.5. 识别

在本文中，我们主要关注人脸验证任务，如前所述。

方法	秩-1准确率	秩-5准确率
MPCNN	88.28	97.66
CryptoFace	92.19 (↑ **3.91%**)	98.44 (↑ **0.78%**)

表3. 1:128封闭集检索。

在3.1节中。然而，CryptoFace可以方便地扩展到人脸识别场景，例如一对多的人脸匹配。在表3中，我们报告了在FHE下进行的1:128闭集排序检索任务的实验结果，使用了随机选择的包含128对图像的LFW子集。延迟差异仅来自特征提取阶段，如之前所述。我们使用Rank-1和Rank-5准确率作为指标来评估性能。实验结果证实，CryptoFace也适用于人脸识别任务。

5. 结论

本文介绍了CryptoFace，这是第一个使用全同态加密（FHE）的端到端加密人脸识别系统。一旦人脸图像被加密，所有后续操作（如特征提取、匹配和分数阈值化）都在加密域内执行，无需解密。CryptoFace背后的关键思想是CryptoFaceNet，这是一种新型架构，它结合了为FHE兼容性优化并减轻加密推理陡峭计算负担的浅层补丁卷积神经网络。在标准人脸识别基准上的实验结果表明，CryptoFace比SoTA FHENets更快，同时实现了更好的验证性能。CryptoFace可以有效地处理高分辨率加密人脸图像，通过提高 $+2\%$的验证准确率，同时保持近分辨率无关的延迟。CryptoFace将促进在需要严格隐私和安全保证的应用中部署安全的人脸识别系统。

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